

Crossover Disambiguation in Kuramoto Sequential Memory

Learning rules for branch-ambiguous retrieval in oscillator dense associative memory

Outline

- 1 Introduction and Motivation
- 2 Literature Review
- 3 Model and Learning Rules
- 4 Numerical Experiments and Results
- 5 Theoretical Analysis
- 6 Conclusions and Outlook

Sequential retrieval in associative memory

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The crossover problem

Overlapping sequences share a state; the transition rule is symmetric in the branches



Objectives



Model overview

Kuramoto dense associative memory and the energetic formulation



Foundations: associative and dense memory

- Krotov & Hopfield — higher-order interactions, dense capacity
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Sequential memory: asymmetric couplings and transition matrices



Open problem: crossover disambiguation



Motivation for Kuramoto dynamics

Smooth retrieval trajectories and physically realizable coupling

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Baseline learning rule

Sequence retrieval algorithm



Single-sequence retrieval under the baseline rule

Failure of the baseline rule at a crossover

Symmetry of the energy landscape



Heterogeneous lag-context learning rules



Benchmark design

Crossover geometries and scoring criteria



Catalogue of learning rules

Rule	Gate

Experimental conditions and parameters



Crossover resolution across the rule catalogue



Superposition of learning rules



Robustness: corruption rate and network size

Discussion: rule performance and limitations



Tensor formulation: symmetric and asymmetric components



Scalability and capacity



Conclusions



Applications

Oscillator-based hardware and neurobiological sequence memory

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Outlook and open questions

